

Package ‘NVIcheckmate’

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Title Extension of checkmate with argument checking adapted for NVIverse

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Description Extends `checkmate` with functions for argument checking that are adapted for NVIverse. NVIcheckmate is intended to be used together with `checkmate`.

URL <https://github.com/NorwegianVeterinaryInstitute/NVIcheckmate>

BugReports <https://github.com/NorwegianVeterinaryInstitute/NVIcheckmate/issues>

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Imports checkmate (>= 2.1.0),
DBI,
keyring

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RoxygenNote 7.3.3

Suggests covr,
desc,
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knitr,
R.rsp,
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roxygen2,
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RODBC,
rstudioapi,
testthat,
NVIpackager,
NVIrpackages

Remotes NorwegianVeterinaryInstitute/NVIpackager,
NorwegianVeterinaryInstitute/NVIrpackages,

Collate 'NVIcheckmate-package.R'
'assert.R'
'makeAssertionFunction.R'
'assert_character.R'

```
'assert_data_frame.R'
'assert_disjunct.R'
'assert_integer.R'
'assert_integerish.R'
'assert_names.R'
'check_choice_character.R'
'check_credentials.R'
'makeWarningFunction.R'
'check_duplicate_rows.R'
'check_non_null.R'
'check_odbc_channel.R'
'check_package.R'
'check_subset_character.R'
'helper.R'
'match_arg.R'
```

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assert	<i>Combine multiple checks into one assertion</i>
--------	---

Description

You can call this function with an arbitrary number of of check* functions, i.e. functions provided by the packages checkmate, NVIcheckmate or your own functions which return TRUE on success and the error message as character(1) otherwise.

Usage

```
assert(..., combine = "or", .var.name = NULL, comment = NULL, add = NULL)
```

Arguments

...	[any] List of calls to check functions.
combine	[character(1)] “or” or “and” to combine the check functions with an OR or AND, respectively.
.var.name	[character(1)] Name of the checked object to print in error messages. Defaults to the heuristic implemented in vname .
comment	[character(1)] Extra information to be appended to the standard error message in assertions.
add	[AssertCollection] Collection to store assertions. See AssertCollection .

Details

The resulting assertion is successful, if combine is “or” (default) and at least one check evaluates to TRUE or combine is “and” and all checks evaluate to TRUE. Otherwise, `assert` throws an informative error message.

Value

Throws an error (or pushes the error message to an [AssertCollection](#) if `add` is not NULL) if the checks fail and invisibly returns TRUE otherwise.

Examples

```
x = 1:10
assert(checkmate::checkNull(x), checkmate::checkInteger(x, any.missing = FALSE))
collection <- checkmate::makeAssertCollection()
assert(checkmate::checkChoice(x, c("a", "b")), checkmate::checkDataFrame(x), add = collection)
collection$getMessages()
```

assert_character	<i>Check if an argument is a vector of type character</i>
------------------	---

Description

Performs assertions if an argument is a vector of type character.

Usage

```
assert_character(
  x,
  n.chars = NULL,
  min.chars = NULL,
  max.chars = NULL,
  pattern = NULL,
  fixed = NULL,
  ignore.case = FALSE,
  any.missing = TRUE,
```

```

    all.missing = TRUE,
    len = NULL,
    min.len = NULL,
    max.len = NULL,
    unique = FALSE,
    sorted = FALSE,
    names = NULL,
    null.ok = FALSE,
    .var.name = checkmate::vname(x),
    comment = NULL,
    add = NULL
  )

```

Arguments

x	[any] Object to check.
n.chars	[integer(1)] Exact number of characters for each element of x.
min.chars	[integer(1)] Minimum number of characters for each element of x.
max.chars	[integer(1)] Maximum number of characters for each element of x.
pattern	[character(1L)] Regular expression as used in grepl . All non-missing elements of x must comply to this pattern.
fixed	[character(1)] Substring to detect in x. Will be used as pattern in grepl with option fixed set to TRUE. All non-missing elements of x must contain this substring.
ignore.case	[logical(1)] See grepl . Default is FALSE.
any.missing	[logical(1)] Are vectors with missing values allowed? Default is TRUE.
all.missing	[logical(1)] Are vectors with no non-missing values allowed? Default is TRUE. Note that empty vectors do not have non-missing values.
len	[integer(1)] Exact expected length of x.
min.len	[integer(1)] Minimal length of x.
max.len	[integer(1)] Maximal length of x.
unique	[logical(1)] Must all values be unique? Default is FALSE.
sorted	[logical(1)] Elements must be sorted in ascending order. Missing values are ignored.
names	[character(1)] Check for names. See check_named for possible values. Default is “any” which performs no check at all. Note that you can use check_subset to check for a specific set of names.

null.ok	[logical(1)] If set to TRUE, x may also be NULL. In this case only a type check of x is performed, all additional checks are disabled.
.var.name	[character(1)] Name of the checked object to print in error messages. Defaults to the heuristic implemented in vname .
comment	[character(1)] Extra information to be appended to the standard error message in assertions.
add	[AssertCollection] Collection to store assertions. See AssertCollection .

Details

The assertions are based on `checkmate::checkCharacter`. `NVCheckmate::assert_character` differs from `checkmate::assert_character` in including the argument `comment = .`. The help is updated to reflect the changes.

This function does not distinguish between NA, NA_integer_, NA_real_, NA_complex_, NA_character_ and NaN.

Value

If the check is successful, the function `assert_character` return x invisibly.
If the check is not successful, `assert_character` throws an error message.

assert_data_frame	<i>Check if an argument is vector of type data frame</i>
-------------------	--

Description

Check if an argument is vector of type data frame.

Usage

```
assert_data_frame(
  x,
  types = character(0L),
  any.missing = TRUE,
  all.missing = TRUE,
  min.rows = NULL,
  max.rows = NULL,
  min.cols = NULL,
  max.cols = NULL,
  nrows = NULL,
  ncols = NULL,
  row.names = NULL,
  col.names = NULL,
  null.ok = FALSE,
  .var.name = checkmate::vname(x),
  comment = NULL,
  add = NULL
)
```

Arguments

<code>x</code>	[any] Object to check.
<code>types</code>	[character] Character vector of class names. Each list element must inherit from at least one of the provided types. The types “logical”, “integer”, “integerish”, “double”, “numeric”, “complex”, “character”, “factor”, “atomic”, “vector”, “atomicvector”, “array”, “matrix”, “list”, “function”, “environment” and “null” are supported. For other types inherits is used as a fallback to check x’s inheritance. Defaults to character(0) (no check).
<code>any.missing</code>	[logical(1)] Are vectors with missing values allowed? Default is TRUE.
<code>all.missing</code>	[logical(1)] Are vectors with no non-missing values allowed? Default is TRUE. Note that empty vectors do not have non-missing values.
<code>min.rows</code>	[integer(1)] Minimum number of rows.
<code>max.rows</code>	[integer(1)] Maximum number of rows.
<code>min.cols</code>	[integer(1)] Minimum number of columns.
<code>max.cols</code>	[integer(1)] Maximum number of columns.
<code>nrows</code>	[integer(1)] Exact number of rows.
<code>ncols</code>	[integer(1)] Exact number of columns.
<code>row.names</code>	[character(1)] Check for row names. Default is “NULL” (no check). See <code>checkmate::check_named</code> for possible values. Note that you can use <code>checkmate::check_subset</code> to check for a specific set of names.
<code>col.names</code>	[character(1)] Check for column names. Default is “NULL” (no check). See <code>checkmate::check_named</code> for possible values. Note that you can use <code>checkmate::check_subset</code> to test for a specific set of names.
<code>null.ok</code>	[logical(1)] If set to TRUE, x may also be NULL. In this case only a type check of x is performed, all additional checks are disabled.
<code>.var.name</code>	[character(1)] Name of the checked object to print in error messages. Defaults to the heuristic implemented in vname .
<code>comment</code>	[character(1)] Extra information to be appended to the standard error message in assertions.
<code>add</code>	[AssertCollection] Collection to store assertions. See AssertCollection .

Details

The assertions are based on `checkmate::check_data_frame`. `NVcheckmate::assert_data_frame` differs from `checkmate::assert_data_frame` in including the argument comment. The help is updated to reflect the changes.

This function does not distinguish between NA, NA_integer_, NA_real_, NA_complex_, NA_character_ and NaN.

Value

Depending on the function prefix:

If the check is successful, the function `assert_Integer` return x invisibly, whereas `check_Integer` return TRUE.

If the check is not successful, `assert_Integer` throws an error message and `check_Integer` returns a string with the error message.

If the check is successful, the function `assert_data_frame` return x invisibly.

If the check is not successful, `assert_data_frame` throws an error message.

assert_disjunct	<i>Check if an argument is disjunct from a given set</i>
-----------------	--

Description

Check if an argument is disjunct from a given set.

Usage

```
assert_disjunct(
  x,
  y,
  fmatch = FALSE,
  .var.name = checkmate::vname(x),
  comment = NULL,
  add = NULL
)
```

Arguments

x	[any] Object to check.
y	[atomic] Other set.
fmatch	[logical(1)] Use the set operations implemented in <code>fmatch</code> in package fastmatch . If fastmatch is not installed, this silently falls back to <code>match</code> . <code>fastmatch::fmatch</code> modifies y by reference: A hash table is added as attribute which is used in subsequent calls.
.var.name	[character(1)] Name of the checked object to print in error messages. Defaults to the heuristic implemented in <code>vname</code> .

comment	[character(1)] Extra information to be appended to the standard error message in assertions.
add	[AssertCollection] Collection to store assertions. See AssertCollection .

Details

The assertions are based on `checkmate::checkDisjunct`. `NVcheckmate::assert_disjunct` differs from `checkmate::assert_disjunct` in including the argument `comment = .`. The help is updated to reflect the changes.

Value

If the check is successful, the function `assert_disjunct` return `x` invisibly.
If the check is not successful, `assert_disjunct` throws an error message.

<code>assert_integer</code>	<i>Check if an argument is vector of type integer</i>
-----------------------------	---

Description

Check if an argument is vector of type integer.

Usage

```
assert_integer(
  x,
  lower = -Inf,
  upper = Inf,
  any.missing = TRUE,
  all.missing = TRUE,
  len = NULL,
  min.len = NULL,
  max.len = NULL,
  unique = FALSE,
  sorted = FALSE,
  names = NULL,
  typed.missing = FALSE,
  null.ok = FALSE,
  .var.name = checkmate::vname(x),
  comment = NULL,
  add = NULL
)
```

Arguments

x	[any] Object to check.
lower	[numeric(1)] Lower value all elements of x must be greater than or equal to.

upper	[numeric(1)] Upper value all elements of x must be lower than or equal to.
any.missing	[logical(1)] Are vectors with missing values allowed? Default is TRUE.
all.missing	[logical(1)] Are vectors with no non-missing values allowed? Default is TRUE. Note that empty vectors do not have non-missing values.
len	[integer(1)] Exact expected length of x.
min.len	[integer(1)] Minimal length of x.
max.len	[integer(1)] Maximal length of x.
unique	[logical(1)] Must all values be unique? Default is FALSE.
sorted	[logical(1)] Elements must be sorted in ascending order. Missing values are ignored.
names	[character(1)] Check for names. See check_named for possible values. Default is “any” which performs no check at all. Note that you can use check_subset to check for a specific set of names.
typed.missing	[logical(1)] If set to FALSE (default), all types of missing values (NA, NA_integer_, NA_real_, NA_character_ or NA_character_) as well as empty vectors are allowed while type-checking atomic input. Set to TRUE to enable strict type checking.
null.ok	[logical(1)] If set to TRUE, x may also be NULL. In this case only a type check of x is performed, all additional checks are disabled.
.var.name	[character(1)] Name of the checked object to print in error messages. Defaults to the heuristic implemented in vname .
comment	[character(1)] Extra information to be appended to the standard error message in assertions.
add	[AssertCollection] Collection to store assertions. See AssertCollection .

Details

The assertions are based on `checkmate::checkInteger`. `NVCheckmate::assert_integer` differs from `checkmate::assert_integer` in including the argument `comment = .`. The help is updated to reflect the changes.

This function does not distinguish between NA, NA_integer_, NA_real_, NA_complex_, NA_character_ and NaN.

Value

Depending on the function prefix:

If the check is successful, the function `assert_Integer` return x invisibly, whereas `check_Integer` return TRUE.

If the check is not successful, `assert_Integer` throws an error message and `check_Integer` returns a string with the error message.

If the check is successful, the function `assert_integer` return `x` invisibly.

If the check is not successful, `assert_integer` throws an error message.

<code>assert_integerish</code>	<i>Check if an argument is vector of type integer</i>
--------------------------------	---

Description

Check if an argument is vector of type integer.

Usage

```
assert_integerish(
  x,
  tol = sqrt(.Machine$double.eps),
  lower = -Inf,
  upper = Inf,
  any.missing = TRUE,
  all.missing = TRUE,
  len = NULL,
  min.len = NULL,
  max.len = NULL,
  unique = FALSE,
  sorted = FALSE,
  names = NULL,
  typed.missing = FALSE,
  null.ok = FALSE,
  .var.name = checkmate::vname(x),
  comment = NULL,
  add = NULL
)
```

Arguments

<code>x</code>	[any] Object to check.
<code>tol</code>	[double(1)] Numerical tolerance used to check whether a double or complex can be converted. Default is <code>sqrt(.Machine\$double.eps)</code> .
<code>lower</code>	[numeric(1)] Lower value all elements of <code>x</code> must be greater than or equal to.
<code>upper</code>	[numeric(1)] Upper value all elements of <code>x</code> must be lower than or equal to.
<code>any.missing</code>	[logical(1)] Are vectors with missing values allowed? Default is TRUE.
<code>all.missing</code>	[logical(1)] Are vectors with no non-missing values allowed? Default is TRUE. Note that empty vectors do not have non-missing values.

len	[integer(1)] Exact expected length of x.
min.len	[integer(1)] Minimal length of x.
max.len	[integer(1)] Maximal length of x.
unique	[logical(1)] Must all values be unique? Default is FALSE.
sorted	[logical(1)] Elements must be sorted in ascending order. Missing values are ignored.
names	[character(1)] Check for names. See check_named for possible values. Default is “any” which performs no check at all. Note that you can use check_subset to check for a specific set of names.
typed.missing	[logical(1)] If set to FALSE (default), all types of missing values (NA, NA_integer_, NA_real_, NA_character_ or NA_character_) as well as empty vectors are allowed while type-checking atomic input. Set to TRUE to enable strict type checking.
null.ok	[logical(1)] If set to TRUE, x may also be NULL. In this case only a type check of x is performed, all additional checks are disabled.
.var.name	[character(1)] Name of the checked object to print in error messages. Defaults to the heuristic implemented in vname .
comment	[character(1)] Extra information to be appended to the standard error message in assertions.
add	[AssertCollection] Collection to store assertions. See AssertCollection .

Details

The assertions are based on `checkmate::checkIntegerish`. `NVcheckmate::assert_integerish` differs from `checkmate::assert_integerish` in including the argument `comment =`. The help is updated to reflect the changes.

This function does not distinguish between NA, NA_integer_, NA_real_, NA_complex_, NA_character_ and NaN.

Value

Depending on the function prefix:

If the check is successful, the function `assert_Integerish` return x invisibly, whereas `check_Integerish` return TRUE.

If the check is not successful, `assert_Integerish` throws an error message and `check_Integerish` returns a string with the error message.

If the check is successful, the function `assert_integerish` return x invisibly.

If the check is not successful, `assert_integerish` throws an error message.

assert_names

*Check names to comply to specific rules***Description**

Performs assertions with various checks on character vectors, usually names.

Usage

```
assert_names(
  x,
  type = "named",
  subset.of = NULL,
  must.include = NULL,
  permutation.of = NULL,
  identical.to = NULL,
  disjunct.from = NULL,
  what = "names",
  .var.name = checkmate::vname(x),
  comment = NULL,
  add = NULL
)
```

Arguments

x	[character NULL] Names to check using rules defined via type.
type	[character(1)] Type of formal check(s) to perform on the names. unnamed: Checks x to be NULL. named: Checks x for regular names which excludes names to be NA or empty (""). unique: Performs checks like with “named” and additionally tests for non-duplicated names. strict: Performs checks like with “unique” and additionally fails for names with UTF-8 characters and names which do not comply to R’s variable name restrictions. As regular expression, this is “ <code>^[.\\]*\\[a-zA-Z\\]+\\[a-zA-Z0-9._]*\$</code> ”. ids: Same as “strict”, but does not enforce uniqueness. Note that for zero-length x, all these name checks evaluate to TRUE.
subset.of	[character] Names provided in x must be subset of the set subset.of.
must.include	[character] Names provided in x must be a superset of the set must.include.
permutation.of	[character] Names provided in x must be a permutation of the set permutation.of. Duplicated names in permutation.of are stripped out and duplicated names in x thus lead to a failed check. Use this argument instead of identical.to if the order of the names is not relevant.

identical.to	[character] Names provided in x must be identical to the vector identical.to. Use this argument instead of permutation.of if the order of the names is relevant.
disjunct.from	[character] Names provided in x must may not be present in the vector disjunct.from.
what	[character(1)] Type of name vector to check, e.g. “names” (default), “colnames” or “row-names”.
.var.name	[character(1)] Name of the checked object to print in error messages. Defaults to the heuristic implemented in vname .
comment	[character(1)] Extra information to be appended to the standard error message in assertions.
add	[AssertCollection] Collection to store assertions. See AssertCollection .

Details

The assertions are based on `checkmate::checkNames`. `NVcheckmate::assert_names` differs from `checkmate::assert_names` in including the argument `comment = .`. The help is updated to reflect the changes.

Value

If the check is successful, the function `assert_names` return x invisibly.
If the check is not successful, `assert_names` throws an error message.

check_choice_character

Check if an object is an element of a given set

Description

Check if an object is an element of a given set in the object name. The function is based on `checkmate::check_choice`, but includes the argument `ignore.case`.

Usage

```
check_choice_character(
  x,
  choices,
  null.ok = FALSE,
  ignore.case = FALSE,
  fmatch = FALSE
)

assert_choice_character(
  x,
  choices,
  null.ok = FALSE,
```

```

    ignore.case = FALSE,
    fmatch = FALSE,
    .var.name = checkmate::vname(x),
    comment = NULL,
    add = NULL
  )

```

Arguments

x	[any] Object to check.
choices	[character] Set of possible values.
null.ok	[logical(1)] If set to TRUE, x may also be NULL. In this case only a type check of x is performed, all additional checks are disabled.
ignore.case	[logical(1)] Case is ignored if TRUE. Default is FALSE.
fmatch	[logical(1)] Use the set operations implemented in fmatch in package fastmatch . If fastmatch is not installed, this silently falls back to match . <code>fastmatch::fmatch</code> modifies y by reference: A hash table is added as attribute which is used in subsequent calls.
.var.name	[character(1)] Name of the checked object to print in error messages. Defaults to the heuristic implemented in vname .
comment	[character(1)] Extra information to be appended to the standard error message in assertions.
add	[AssertCollection] Collection to store assertions. See AssertCollection .

Details

The object must be of type character. The check is intended for functions were using camelCase may make the argument easier to remember. Therefore, the function will ignore case if `ignore.case = TRUE`.

Value

Depending on the function prefix:

If the check is successful, the function `assert_choice_character` return x invisibly, whereas `check_choice_character` return TRUE.

If the check is not successful, `assert_choice_character` throws an error message and `check_choice_character` returns a string with the error message.

Author(s)

Petter Hopp Petter.Hopp@vetinst.no

Examples

```
# returns TRUE
check_choice_character(x = "APPLE",
                      choices = c("Apple", "Pear", "Orange"),
                      null.ok = FALSE,
                      ignore.case = TRUE,
                      fmatch = FALSE)
```

check_credentials	<i>Check if an object is a service for which credentials are saved in the user profile</i>
-------------------	--

Description

Check if credentials are saved in the user profile for the service.

Usage

```
check_credentials(x)

assert_credentials(
  x,
  .var.name = checkmate::vname(x),
  comment = NULL,
  add = NULL
)
```

Arguments

x	[any] Object to check.
.var.name	[character(1)] Name of the checked object to print in error messages. Defaults to the heuristic implemented in vname .
comment	[character(1)] Extra information to be appended to the standard error message in assertions.
add	[AssertCollection] Collection to store assertions. See AssertCollection .

Value

Depending on the function prefix:

If the check is successful, the function `assert_credentials` return `x` invisibly, whereas `check_credentials` return `TRUE`.

If the check is not successful, `assert_credentials` throws an error message and `check_credentials` returns a string with the error message.

Examples

```
check_credentials(x = "PJS")
```

check_duplicate_rows *Check if data has duplicate rows*

Description

Check if data has duplicate rows for one or more group variables.

Usage

```
check_duplicate_rows(x, group_var)

assert_duplicate_rows(
  x,
  group_var,
  .var.name = checkmate::vname(x),
  comment = NULL,
  add = NULL
)

warn_duplicate_rows(
  x,
  group_var,
  .var.name = checkmate::vname(x),
  comment = NULL,
  add = NULL
)
```

Arguments

x	[any] Object to check.
group_var	[character] The variables in the data that should constitute an unique id. Defaults to NULL.
.var.name	[character(1)] Name of the checked object to print in error messages. Defaults to the heuristic implemented in vname .
comment	[character(1)] Extra information to be appended to the standard error message in assertions.
add	[AssertCollection] Collection to store assertions. See AssertCollection .

Details

Used the check that the group variables can be a unique identifier of rows in the data. The data can be data.frame, tibble or data.table.

The warn_duplicate_rows is mainly intended to be used to check data in routine scripts transforming data, but in a way that produce a warning instead of an error to avoid that the script stop running. In particular, it is intended to check data after steps where id's may have been duplicated. This can be important to avoid problems with joins/mergings in consecutive steps.

Value

Depending on the function prefix:

If the check is successful, the function `assert_duplicate_rows` return `x` invisibly, whereas `warn_duplicate_rows` and `check_duplicate_rows` return `TRUE`.

If the check is not successful, `assert_duplicate_rows` throws an error message, `warn_duplicate_rows` throws a warning message, and `check_duplicate_rows` returns a string with the error/warning message.

Author(s)

Petter Hopp Petter.Hopp@vetinst.no

Examples

```
## Not run:
# Creating data
pt_data <- as.data.frame(list("prodnr8" = c("01234567", "12345678"), "P157" = c(200, 300)))
prod_name <- as.data.frame(
  list("prodnr8" = c("01234567", "12345678", "12345678"),
       "prodnr10" = c("0123456713", "1234567814", "1234567826"),
       "navn" = c("Ola Nordmann", "Kari Trestakk", "Per Trestakk")))
# merge
pt_data <- merge(pt_data, prod_name, by = "prodnr8")
# checking for double rows per prodnr8
assert_duplicate_rows(x = pt_data,
                      group_var = "prodnr8",
                      comment = "Checked after merging pt_data med prod_name")

## End(Not run)
```

check_non_null

Check if two or more arguments are NULL or NA

Description

Check whether two or more arguments that may replace each other are NULL or NA.

Usage

```
check_non_null(x)
```

```
assert_non_null(x, .var.name = checkmate::vname(x), comment = NULL, add = NULL)
```

Arguments

<code>x</code>	[any] List with objects to check.
<code>.var.name</code>	[character(1)] Name of the checked object to print in error messages. Defaults to the heuristic implemented in vname .

comment	[character(1)] Extra information to be appended to the standard error message in assertions.
add	[AssertCollection] Collection to store assertions. See AssertCollection .

Details

In some functions, one argument may replace another. The check is used to ensure that at least one of the arguments have a value. Specific checks for each argument should be used in addition.

Value

Depending on the function prefix:

If the check is successful, the function `assert_non_null` return `x` invisibly, whereas `check_non_null` return `TRUE`.

If the check is not successful, `assert_non_null` throws an error message and `check_non_null` returns a string with the error message.

Examples

```
data <- NULL
nrows_in_data <- 56
check_non_null(x = list(data, nrows_in_data))
```

check_odbc_channel	<i>Check if an object is an open ODBC-channel</i>
--------------------	---

Description

Check if an object is an open ODBC-channel.

Usage

```
check_odbc_channel(x, dbservice = NULL, dbinterface = "odbc")

assert_odbc_channel(
  x,
  dbservice = NULL,
  dbinterface = "odbc",
  .var.name = checkmate::vname(x),
  comment = NULL,
  add = NULL
)
```

Arguments

x	[any] Object to check.
dbservice	[character(1)] The database the channel should be connect to.

dbinterface	[character(1)] The R-package that is used for interface towards the data base. Defaults to NULL.
.var.name	[character(1)] Name of the checked object to print in error messages. Defaults to the heuristic implemented in vname .
comment	[character(1)] Extra information to be appended to the standard error message in assertions.
add	[AssertCollection] Collection to store assertions. See AssertCollection .

Details

The check uses `RODBC::odbcValidChannel` to check if the argument is an open ODBC-channel. Be aware that this function is an internal RODBC-function and may change within that package without warning.

Value

Depending on the function prefix:

If the check is successful, the function `assert_odbc_channel` return x invisibly, whereas `check_odbc_channel` return TRUE.

If the check is not successful, `assert_odbc_channel` throws an error message and `check_odbc_channel` returns a string with the error message.

Author(s)

Petter Hopp Petter.Hopp@vetinst.no

Examples

```
## Not run:
# login to PJS
journal_rapp <- login("PJS")

# error check:
check_odbc_channel(x = "journal_rapp", dbservice = "PJS")

## End(Not run)
```

check_package	<i>Check if an object is an installed or attached package</i>
---------------	---

Description

Check if a package is installed and if it is installed, check if the version is equal to or higher than the required version or check if a package is attached.

Usage

```

check_package(x, version = NULL, type = "installed")

assert_package(
  x,
  version = NULL,
  type = "installed",
  .var.name = checkmate::vname(x),
  comment = NULL,
  add = NULL
)

```

Arguments

x	[any] Object to check.
version	[character(1)] The required version of the installed package. May be NULL.
type	[character(1)] Type of formal check(s) to perform on the package. installed: Checks if x is installed. attached: Checks if x is attached.
.var.name	[character(1)] Name of the checked object to print in error messages. Defaults to the heuristic implemented in vname .
comment	[character(1)] Extra information to be appended to the standard error message in assertions.
add	[AssertCollection] Collection to store assertions. See AssertCollection .

Value

Depending on the function prefix:

If the check is successful, the function `assert_package` return x invisibly, whereas `check_package` return TRUE.

If the check is not successful, `assert_package` throws an error message and `check_package` returns a string with the error message.

Examples

```

# returns TRUE i.e. no error message
check_package(x = "checkmate", version = "2.0.0", type = "installed")

# returns an error message
check_package(x = "nopackage", type = "installed")
check_package(x = "nopackage", type = "attached")

```

check_subset_character

Check if an argument is a subset of a given set

Description

Check if an object is a subset of a given set in the object name. The function is based on `checkmate::check_subset`, but includes the argument `ignore.case`.

Usage

```
check_subset_character(
  x,
  choices,
  ignore.case = FALSE,
  empty.ok = TRUE,
  fmatch = FALSE
)

assert_subset_character(
  x,
  choices,
  ignore.case = FALSE,
  empty.ok = TRUE,
  fmatch = FALSE,
  .var.name = checkmate::vname(x),
  comment = NULL,
  add = NULL
)
```

Arguments

<code>x</code>	[any] Object to check.
<code>choices</code>	[character] Set of possible values. May be empty.
<code>ignore.case</code>	[logical(1)] Case is ignored if TRUE. Default is FALSE.
<code>empty.ok</code>	[logical(1)] Treat zero-length <code>x</code> as subset of any set <code>choices</code> (this includes NULL)? Default is TRUE.
<code>fmatch</code>	[logical(1)] Use the set operations implemented in <code>fmatch</code> in package fastmatch . If fastmatch is not installed, this silently falls back to <code>match</code> . <code>fastmatch::fmatch</code> modifies <code>y</code> by reference: A hash table is added as attribute which is used in subsequent calls.
<code>.var.name</code>	[character(1)] Name of the checked object to print in error messages. Defaults to the heuristic implemented in <code>vname</code> .

comment	[character(1)] Extra information to be appended to the standard error message in assertions.
add	[AssertCollection] Collection to store assertions. See AssertCollection .

Details

The object must be of type character. The check is intended for functions were using camelCase may make the argument easier to remember. Therefore, the function will ignore case if `ignore.case = TRUE`.

Value

Depending on the function prefix:

If the check is successful, the function `assert_subset_character` return `x` invisibly, whereas `check_subset_character` return `TRUE`.

If the check is not successful, `assert_subset_character` throws an error message and `check_subset_character` returns a string with the error message.

Author(s)

Petter Hopp Petter.Hopp@vetinst.no

Examples

```
# returns TRUE
check_subset_character(x = "Apple",
                      choices = c("apple", "pear", "orange", "banana"),
                      ignore.case = TRUE)

check_subset_character(x = c("Apple", "Pear"),
                      choices = c("apple", "pear", "orange", "banana"),
                      ignore.case = TRUE)

# returns a message
check_subset_character(x = "Tomato",
                      choices = c("apple", "pear", "orange", "banana"),
                      ignore.case = TRUE)
```

makeAssertionFunction *Turn a Check into an Assertion*

Description

`makeAssertionFunction` can be used to automatically create an assertion function based on a check function (see example). This is a modification of `checkmate::makeAssertionFunction` that includes the argument `comment` in the assertion function.

Usage

```
makeAssertionFunction(
  check.fun,
  c.fun = NULL,
  coerce = FALSE,
  env = parent.frame()
)
```

Arguments

check.fun	[function] Function which checks the input. Must return TRUE on success and a string with the error message otherwise.
c.fun	[character(1)] If not NULL, instead of calling the function check.fun, use .Call to call a C function “c.fun” with the identical set of parameters. The C function must be registered as a native symbol, see CallExternal . Useful if check.fun is just a simple wrapper.
coerce	[logical(1)] If TRUE, injects some lines of code to convert numeric values to integer after an successful assertion. Currently used in assert_count , assert_int and assert_integerish .
env	[environment] The environment of the created function. Default is the parent.frame, see sys.parent .

Details

The code is imported from checkmate. The modifications in the code is marked. The argument use.namespace is deleted as checkmate::makeAssertion and checkmate::vname always should be used.

makeWarning

Turn a Check into an Assertion

Description

makeAssertionFunction can be used to automatically create an assertion function based on a check function (see example). This is a modification of checkmate::makeAssertionFunction that includes the argument comment in the assertion function.

Usage

```
makeWarning(x, res, var.name)
```

```
makeWarningFunction(
  check.fun,
  c.fun = NULL,
  coerce = FALSE,
  env = parent.frame()
)
```

Arguments

<code>x</code>	[any] Object to check.
<code>res</code>	[TRUE character(1)] The result of a check function: TRUE for successful checks, and an error message as string otherwise.
<code>var.name</code>	[character(1)] The custom name for <code>x</code> as passed to any <code>assert*</code> function. Defaults to a heuristic name lookup.
<code>check.fun</code>	[function] Function which checks the input. Must return TRUE on success and a string with the error message otherwise.
<code>c.fun</code>	[character(1)] If not NULL, instead of calling the function <code>check.fun</code> , use <code>.Call</code> to call a C function “ <code>c.fun</code> ” with the identical set of parameters. The C function must be registered as a native symbol, see CallExternal . Useful if <code>check.fun</code> is just a simple wrapper.
<code>coerce</code>	[logical(1)] If TRUE, injects some lines of code to convert numeric values to integer after an successful assertion. Currently used in assert_count , assert_int and assert_integerish .
<code>env</code>	[environment] The environment of the created function. Default is the <code>parent.frame</code> , see sys.parent .

Details

The code is imported from `checkmate`. The modifications in the code is marked. The argument `use.namespace` is deleted as `checkmate::makeAssertion` and `checkmate::vname` always should be used.

Value

`makeAssertion` invisibly returns the checked object if the check was successful, and an exception is raised (or its message stored in the collection) otherwise. `makeAssertionFunction` returns a function.

Examples

```
## Not run:
# Simple custom check function
checkFalse = function(x) if (!identical(x, FALSE)) "Must be FALSE" else TRUE

# Create the respective assert function
assertFalse = function(x, .var.name = vname(x), add = NULL) {
  res = checkFalse(x)
  makeAssertion(x, res, .var.name, add)
}

# Alternative: Automatically create such a function
assertFalse = makeAssertionFunction(checkFalse)
print(assertFalse)

## End(Not run)
```


Description

match_arg matches arguments against a table of candidate values as specified by choices. Matching is done using [grep](#), and arguments may be abbreviated. If ignore.case = TRUE, case is ignored when performing the matching.

Usage

```
match_arg(
  x,
  choices,
  several.ok = FALSE,
  ignore.case = FALSE,
  .var.name = checkmate::vname(x),
  comment = NULL,
  add = NULL
)
```

Arguments

x	[character] User provided argument to match.
choices	[character] Candidates to match x with.
several.ok	[logical(1)] If TRUE, x should be allowed to have more than one element.
ignore.case	[logical(1)] Case is ignored if TRUE. Default is FALSE.
.var.name	[character(1)] Name of the checked object to print in error messages. Defaults to the heuristic implemented in vname .
comment	[character(1)] Extra information to be appended to the standard error message in assertions.
add	[AssertCollection] Collection to store assertions. See AssertCollection .

Details

Partial Argument Matching

This is an extensions to [matchArg](#) with support for ignore.case and comment. [matchArg](#) is an extension of [match.arg](#) with support for [AssertCollection](#). The behavior is very similar to [match.arg](#), with a few exceptions: * NULL is not a valid value for x. * When several.ok = TRUE, it is required that all values for x match a value in choices and that each value in x matches only one value in choices. * When several.ok = FALSE, it is required that length(x) == 1 and that x matches one and only one value in choices.

Value

Character vector with subset of choices.

Examples

```
match_arg("a", choices = c("Apple", "Banana"), ignore.case = TRUE)
match_arg(c("a", "Ban"), choices = c("Apple", "Banana"), several.ok = TRUE, ignore.case = TRUE)
```

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